

Model Selection and Evaluation on the Iris Species Dataset

Introduction

Machine learning models are used to identify patterns in data and make predictions. Different algorithms may perform differently depending on the dataset and problem being solved. Therefore, model selection and evaluation are important steps in the machine learning pipeline.

The objective of this project was to compare multiple machine learning classification algorithms on the Iris Species Dataset and determine the most suitable model based on evaluation metrics and cross-validation performance.

Dataset Overview

The Iris dataset is one of the most widely used datasets for machine learning classification tasks. It contains measurements of iris flowers from three different species.

Features

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Target Classes

- Iris Setosa
- Iris Versicolor
- Iris Virginica

Dataset Size

- Total Samples: 150
- Features: 4
- Classes: 3

The dataset is clean, balanced, and contains no missing values, making it ideal for classification experiments.

Problem Statement

The objective is to predict the species of an iris flower using its physical measurements.

This is a **classification problem** because the target variable consists of categorical labels representing flower species.

Data Preparation

The dataset was loaded using Scikit-learn's built-in Iris dataset loader.

The features (X) consisted of the four flower measurements, while the target variable (y) represented the flower species.

The dataset was divided into training and testing sets using an 80:20 split:

- Training Data: 80%
- Testing Data: 20%

A fixed random state of 42 was used to ensure reproducibility of results.

Models Used

Three machine learning classification algorithms were selected for comparison.

1. Logistic Regression

Logistic Regression is a simple and efficient classification algorithm that predicts class probabilities and assigns observations to the most likely category.

Advantages

- Fast and computationally efficient
 - Easy to interpret
 - Works well on linearly separable data
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2. Decision Tree Classifier

Decision Trees classify data using a tree-like structure of decision rules.

Advantages

- Easy to visualize and understand
 - Handles non-linear relationships
 - Requires minimal preprocessing
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3. Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve predictive performance.

Advantages

- High accuracy
 - Reduces overfitting
 - Handles complex patterns effectively
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Evaluation Metrics

The models were evaluated using the following metrics:

Accuracy

Measures the proportion of correctly classified samples.

Precision

Measures how often positive predictions are correct.

Recall

Measures the model's ability to identify all relevant instances.

F1 Score

Represents the balance between Precision and Recall.

Cross Validation Accuracy

Measures model performance across multiple train-test splits and provides a more reliable estimate of generalization performance.

Results

Logistic Regression

- Accuracy: 1.00
 - Precision: 1.00
 - Recall: 1.00
 - F1 Score: 1.00
 - Cross Validation Accuracy: 0.9733 (97.33%)
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Decision Tree

- Accuracy: 1.00
 - Precision: 1.00
 - Recall: 1.00
 - F1 Score: 1.00
 - Cross Validation Accuracy: 0.9533 (95.33%)
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Random Forest

- Accuracy: 1.00
 - Precision: 1.00
 - Recall: 1.00
 - F1 Score: 1.00
 - Cross Validation Accuracy: 0.9667 (96.67%)
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Model Comparison

Model	Accuracy	Precision	Recall	F1 Score	Cross Validation Accuracy
Logistic Regression	1.00	1.00	1.00	1.00	97.33%
Decision Tree	1.00	1.00	1.00	1.00	95.33%
Random Forest	1.00	1.00	1.00	1.00	96.67%

Analysis

All three models achieved perfect Accuracy, Precision, Recall, and F1 Scores on the test dataset. This indicates that each model successfully classified all test samples.

However, relying only on test accuracy can sometimes be misleading because performance may vary depending on the train-test split.

To obtain a more reliable evaluation, 5-fold cross-validation was performed. Cross-validation measures how well a model generalizes to unseen data by repeatedly training and testing the model on different subsets of the dataset.

The cross-validation results showed clear differences among the models:

- Logistic Regression: 97.33%
- Random Forest: 96.67%
- Decision Tree: 95.33%

These results indicate that Logistic Regression generalized slightly better than the other models.

Best Model Selection

Based on the evaluation results, **Logistic Regression** was selected as the best model for this project.

Reasons for selection:

1. Achieved perfect Accuracy, Precision, Recall, and F1 Score.
 2. Obtained the highest Cross Validation Accuracy (97.33%).
 3. Simpler and more interpretable than Random Forest.
 4. Demonstrated strong generalization performance.
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Conclusion

This project compared three machine learning classification algorithms on the Iris Species Dataset: Logistic Regression, Decision Tree, and Random Forest.

Although all models achieved perfect performance on the test dataset, cross-validation revealed differences in their ability to generalize to unseen data. Logistic Regression achieved the highest cross-validation accuracy of 97.33%, making it the most reliable model for this classification task.

The project demonstrates the importance of evaluating multiple models and using cross-validation when selecting the most appropriate machine learning algorithm. Proper model evaluation helps prevent overfitting and ensures that the chosen model performs well on new data.